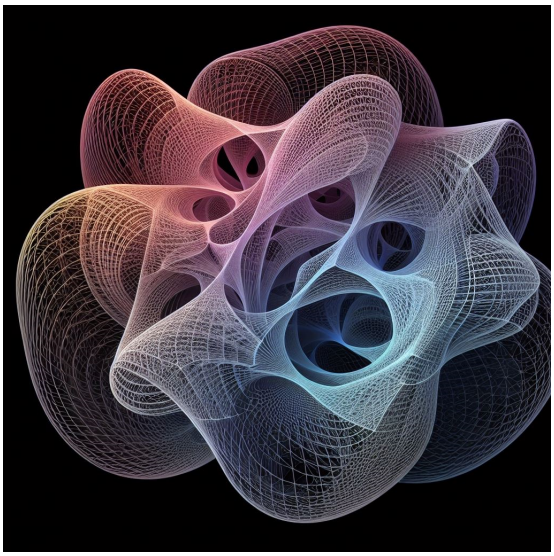

LINMA2474 - High-Dimensional Data Analysis and Optimization

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This summary may not be up-to-date, the newer version is available at this address:
https://github.com/SimonDesmidt/Syntheses

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UCLouvain

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Optimization on manifolds

1.1 Introduction

Classical optimization methods like the gradient descent solve problems of the form

$$\min_{x \in \mathcal{M}} f(x) \tag{1.1}$$

for a set M. The methods rely on two key properties:

- Linearity: x_k and $\nabla f(x_k)$ belong to some vector space, in which they can be combined with linear operations;
- Inner product: $\nabla f(x_k)$ is the unique element of $\mathbb{R}^D$ such that

$$\forall v \in \mathbb{R}^D,\ Df(x)[v] = \langle v, \nabla f(x) \rangle \tag{1.2}$$

  where $Df(x)[v] = \lim_{t \to 0} \frac{f(x+tv)-f(x)}{t}$ is the directional derivative of f at x in the direction v.

There are two ways to see the problem 1.1: as a constrained optimization problem, or as an unconstrained optimization problem assuming that nothing else exists outside the set $\mathcal{M}$. Optimization on manifolds extends the classical unconstrained optimization algorithms to problems whose search space is a manifold (will be defined later).

1.2 Definitions

1.2.1 Definition and properties of a manifold

Definition 1.1 (Optimisation on manifolds). To minimize a function f on a manifold $\mathcal{M}$, we need several objects:

- A local defining function $h : \mathbb{R}^d \to \mathbb{R}$ such that $h^{-1}(0) = \mathcal{M}$;
- The tangent space of $\mathcal{M}$ at some point x is the local linearization of $\mathcal{M}$ at $x \in \mathcal{M}$: $u \in$tangent space of $\mathcal{M}$ at x iff $u \in Ker(Dh(x))$;
- An inner product on the tangent spaces to define a new notion of gradient:

$$Df(x)[v] = \langle \nabla f(x), v \rangle$$

- A retraction function, i.e. a tool that allows to make a step on a manifold in a given tangent direction.

Definition 1.2 (Smoothness). $F : \mathcal{U} \subseteq \mathcal{E} \to \mathcal{V} \subseteq \mathcal{E}'$, with $\mathcal{U}, \mathcal{V}$ open, is said to be smooth if it is $\mathcal{C}^\infty$ on its domain.

Let us explain the concepts needed for our optimization:

Definition 1.3 (Embedded submanifold and local defining function). Let $\mathcal{E}$ be a linear space of dimension d. A non-empty subset $\mathcal{M}$ of $\mathcal{E}$ is a smooth embedded submanifold of $\mathcal{E}$ of dimension n if either:

- $n = d$ and $\mathcal{M}$ is open in $\mathcal{E}$ (open submanifold);
- $n = d - k$ for some $k \geq 1$ and, for each $x \in \mathcal{M}$, there exists a neighbourhood $\mathcal{U}$ of x in $\mathcal{E}$ and a smooth function $h : \mathcal{U} \to \mathbb{R}^k$. In that case,
  - $\mathcal{M} \cap \mathcal{U} = h^{-1}(0) = \{y \in \mathcal{U} : h(y) = 0\}$ and
  - $rank(Dh(x)) = k$.

Such a function h is called a local defining function for $\mathcal{M}$ at x.

Definition 1.4 (Tangent space). Let $\mathcal{M}$ be a subset of $\mathcal{E}$. For all $x \in \mathcal{M}$, define

$$\mathcal{T}_x\mathcal{M} = \{c'(0) \mid c : \mathcal{I} \to \mathcal{M} \text{ is smooth and } c(0) = x\} \tag{1.3}$$

where $\mathcal{I}$ is any open interval containing $t = 0$. That means that v is in $\mathcal{T}_x\mathcal{M}$ iff there exists a smooth curve on $\mathcal{M}$ passing through x with velocity v.

Consider $\mathcal{M}$ an embedded submanifold of $\mathcal{E}$, $x \in \mathcal{M}$ and the set $\mathcal{T}_x\mathcal{M}$.

- If $\mathcal{M}$ is an open submanifold of $\mathcal{E}$, then $\mathcal{T}_x\mathcal{M} = \mathcal{E}$;
- Otherwise, $\mathcal{T}_x\mathcal{M} = Ker(Dh(x))$ with h any local defining function at x.

Definition 1.5 (Tangent bundle). The tangent bundle is the set of all tangent spaces: $\mathcal{T}\mathcal{M} = \{(x, v) : v \in \mathcal{T}_x\mathcal{M}\}$.

Definition 1.6 (Map between manifolds). Let $\mathcal{M}$ and $\mathcal{M}'$ be embedded submanifolds of $\mathcal{E}$ and $\mathcal{E}'$. A map $F : \mathcal{M} \to \mathcal{M}'$ is smooth iff $F = \bar{F}|_{\mathcal{M}}$ where $\bar{F}$ is some smooth map from a neighbourhood of $\mathcal{M}$ in $\mathcal{E}$ to $\mathcal{E}'$.

Definition 1.7 (Differential of a map between manifolds). The differential of $F : \mathcal{M} \to \mathcal{M}'$ at the point $x \in \mathcal{M}$ is the linear map $DF(x) : \mathcal{T}_x\mathcal{M} \to \mathcal{T}_{F(x)}\mathcal{M}'$ defined by

$$DF(x)[v] = \frac{d}{dt}F(c(t))|_{t=0} = (F \circ c)'(0) \tag{1.4}$$

where c is some smooth curve on $\mathcal{M}$ passing through x at $t = 0$ with velocity $v \in \mathcal{T}_x\mathcal{M}$.

→ Note: the definition does not depend on the choice of the curve c: $DF(x) = D\bar{F}(x)|_{\mathcal{T}_x\mathcal{M}}$.

Definition 1.8 (Retraction function). A retraction on a manifold $\mathcal{M}$ is a smooth map $R : \mathcal{TM} \to \mathcal{M} : (x, v) \to R_x(v)$ such that each curve $c(t) = R_x(tv)$ satisfies $c(0) = x$ and $c'(0) = v$.

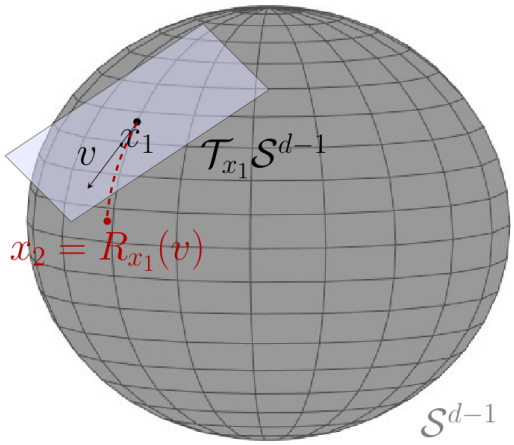

Figure 1.1: Retraction on a sphere.

1.2.2 Riemannian manifolds and metrics

Definition 1.9 (Inner product). As seen in previous courses, an inner product on $\mathcal{T}_x\mathcal{M}$ is a bilinear, symmetric, positive definite function $\langle \cdot, \cdot \rangle_x : \mathcal{T}_x\mathcal{M} \times \mathcal{T}_x\mathcal{M} \to \mathbb{R}$. Note that the inner product depends on the point of linearization. It induces some norm for tangent vectors: $\|u\|_x = \sqrt{\langle u, u \rangle_x}$. A metric on $\mathcal{M}$ is a choice of inner product $\langle \cdot, \cdot \rangle_x$ for each $\mathcal{M}$.

Definition 1.10 (Metric). A metric $x \to \langle \cdot, \cdot \rangle_x$ on $\mathcal{M}$ is a Riemannian metric if it varies smoothly with x, i.e. for all smooth vector fields V, W on $\mathcal{M}$, the function $x \to \langle V(x), W(x) \rangle_x$ is smooth from $\mathcal{M}$ to $\mathbb{R}$.

Definition 1.11 (Riemannian manifold). A Riemannian manifold is a manifold with a Riemannian metric.

Definition 1.12 (Riemannian distance). Let $\mathcal{M}$ be a Riemannian manifold. Given a smooth curve $c : [a, b] \to \mathcal{M}$, we define the length of c as

$$L(c) = \int_a^b \|c'(t)\|_{c(t)} dt \tag{1.5}$$

The Riemannian distance is then defined as $dist(x, y) = \inf_c L(c)$.

Definition 1.13 (Riemannian submanifolds). Let $\mathcal{M}$ be an embedded submanifold of a Euclidean space $\mathcal{E}$. Equipped with the Riemannian metric obtained by restriction of the metric of $\mathcal{E}$, we call $\mathcal{M}$ a Riemannian submanifold of $\mathcal{E}$.

1.2.3 Gradient on manifolds

Definition 1.14 (Riemannian gradient). Let $f : \mathcal{M} \to \mathbb{R}$ be smooth on a Riemannian manifold $\mathcal{M}$. The Riemannian gradient of f is the vector field $\mathrm{grad} f$ on $\mathcal{M}$ uniquely defined by the following identities:

$$\forall (x, v) \in \mathcal{TM}, \qquad Df(x)[v] = \langle v, \mathrm{grad} f(x) \rangle_x \tag{1.6}$$

where $Df(x)$ is the differential of f and $\langle \cdot, \cdot \rangle_x$ is the Riemannian metric.

Theorem 1.15. Let $\mathcal{M}$ be a Riemannian submanifold of $\mathcal{E}$ endowed with the metric $\langle \cdot, \cdot \rangle$ and let $f : \mathcal{M} \to \mathbb{R}$ be a smooth function. The Riemannian gradient of f is given by

$$\mathrm{grad} f(x) = \mathrm{Proj}_x(\nabla \bar{f}(x)) \tag{1.7}$$

where $\bar{f}$ is any smooth extension of f to a neighborhood of $\mathcal{M}$ in $\mathcal{E}$, and $\nabla \bar{f}(x)$ is the Euclidean gradient of $\bar{f}$ at x.

→ Note: for $\mathcal{E} = \mathbb{R}^d$ and using the usual metric $\langle u, v \rangle = u^T v$, the projection operator is

$$\mathrm{Proj}_x(v) = v - (x^T v)x \tag{1.8}$$

Proposition 1.16. Let $f : \mathcal{M}_1 \to \mathcal{M}_2$ and $G : \mathcal{M}_2 \to \mathcal{M}_3$ be smooth, where $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$ are embedded submanifolds of $\mathcal{E}_1, \mathcal{E}_2, \mathcal{E}_3$ respectively. Then

$$G \circ F : \mathcal{M}_1 \to \mathcal{M}_3 : x \to G(F(x)) \tag{1.9}$$

is smooth and the chain rule applies:

$$D(G \circ F)(x)[v] = DG(F(x))[DF(x)[v]] \tag{1.10}$$

1.2.4 Taylor development of functions defined on manifolds

Let $f : \mathcal{M} \to \mathbb{R}$ be smooth and $c : \mathcal{I} \to \mathcal{M}$ be a smooth curve with $c(0) = x$ and $c'(0) = v$, with $\mathcal{I} \subseteq \mathbb{R}$ an open interval around $t = 0$ and $\|v\|_x = 1$. Let us define $g : \mathcal{I} \to \mathbb{R} : t \to g(t) = f(c(t))$. Since $g = f \circ c$ is smooth and maps real numbers to real numbers, it admits a Taylor expansion:

$$g(t) = g(0) + tg'(0) + \mathcal{O}(t^2) \tag{1.11}$$

By the chain rule,

$$g'(t) = Df(c(t))[c'(t)] = \langle \mathrm{grad} f(c(t)), c'(t) \rangle_{c(t)} \tag{1.12}$$

and for $t = 0$,

$$g(0) = f(x) \qquad g'(0) = \langle \mathrm{grad} f(x), v \rangle_x \tag{1.13}$$

Therefore,

$$\begin{aligned} f(c(t)) &= f(x) + t\langle \mathrm{grad} f(x), v \rangle_x + \mathcal{O}(t^2) \\ f(R_x(tv)) &= f(x) + \langle \mathrm{grad} f(x), tv \rangle_x + \mathcal{O}(t^2) \end{aligned} \tag{1.14}$$

And defining $s := tv \in \mathcal{T}_x\mathcal{M}$,

$$f(R_x(s)) = f(x) + \langle \mathrm{grad} f(x), s \rangle_x + \mathcal{O}(\|s\|_x^2) \tag{1.15}$$

This allows to define the Riemannian gradient descent in the next chapter (see section 2.2).

1.3 Geometric considerations

1.3.1 Geodesics

Definition 1.17 (Geodesics). On a Riemannian manifold $\mathcal{M}$, a geodesic is a smooth curve $c : \mathcal{I} \to \mathcal{M}$ such that $c''(t) = 0$ for all $t \in \mathcal{I}$, where $\mathcal{I}$ is an open interval of $\mathbb{R}$. The acceleration $c''(t)$ on a Riemannian submanifold is defined as follows:

$$c''(t) = \mathrm{Proj}_{c(t)}(\ddot{c}(t)) \qquad \text{where} \qquad \ddot{c}(t) := \frac{d^2}{dt^2}c(t) \tag{1.16}$$

For example, the curve $c(t) = \cos(t\|v\|)x + \frac{\sin(t\|v\|)}{\|v\|}v$, where x is the starting point and v the initial velocity ($v \in \mathcal{T}_x\mathcal{S}^{d-1}$), i.e. the curve that traces a great circle on the sphere.

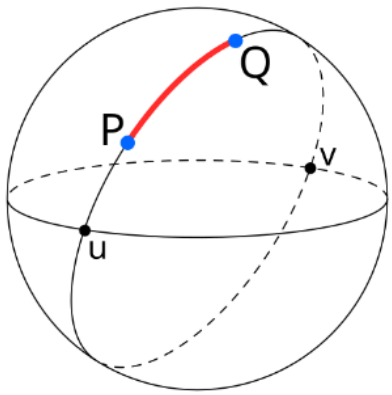

Figure 1.2: Example of geodesic on the sphere.

Theorem 1.18. Every geodesic γ on $\mathcal{M}$ is locally minimizing, i.e. every t in the domain of γ has a neighborhood $\mathcal{I}$ in the domain of γ such that, if $a, b \in \mathcal{I}$ with $a < b$, then the restriction $\gamma|_{[a,b]}$ is a minimizing curve.

→ Note: not all geodesics are minimizing, e.g. on the sphere, two points are always linked by two geodesics, a long one and a short one (red and black curve on figure 1.2).

Theorem 1.19. On a Riemannian manifold, for every $(x, v) \in \mathcal{TM}$, there exists a unique maximal geodesic $\gamma : \mathcal{I} \to \mathcal{M}$ with $\gamma(0) = x$ and $\gamma'(0) = v$, where maximal means that the interval $\mathcal{I}$ is as large as possible. This is equivalent to a Cauchy problem:

$$\gamma''(t) = 0 \qquad \text{such that} \begin{cases} \gamma(0) = x \\ \gamma'(0) = v \end{cases} \tag{1.17}$$

Definition 1.20 (Exponential map). Consider the set

$$\mathcal{D} = \{(x, v) \in \mathcal{TM} \ : \gamma \text{ is defined on an interval containing } [0, 1]\}$$

which is a subset of the tangent bundle. The exponential map $Exp : \mathcal{D} \to \mathcal{M}$ is defined by the relation $Exp(x, v) = \gamma(1)$.

→ Note: the exponential map is a retraction.

1.3.2 Geodesic convexity

As a reminder, in linear space, a set $\mathcal{S}$ is convex if, for all $x, y \in \mathcal{S}$, the line segment $t \to (1-t)x + ty$ for $t \in [0,1]$ is also in $\mathcal{S}$. A function $f : \mathcal{S} \to \mathbb{R}$ is a convex function if $\mathcal{S}$ is convex and for all $x, y \in \mathcal{S}$ we have

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y) \qquad \forall t \in [0,1] \tag{1.18}$$

Definition 1.21 (Geodesic convexity). A subset $\mathcal{S}$ of a Riemannian manifold $\mathcal{M}$ is geodesically convex if, for every $x, y \in \mathcal{S}$, there exists a geodesic segment $c : [0,1] \to \mathcal{M}$ such that $c(0) = x$, $c(1) = y$ and $c(t)$ is in $\mathcal{S}$ for all $t \in [0,1]$.

Definition 1.22 (Geodesically convex function). A function $f : \mathcal{S} \to \mathbb{R}$ is geodesically convex if $\mathcal{S}$ is geodesically convex and $f \circ c : [0,1] \to \mathbb{R}$ is convex for each geodesic segment $c : [0,1] \to \mathcal{M}$ whose image is in $\mathcal{S}$.

Theorem 1.23. Let $\mathcal{S}$ be a geodesically convex set on a Riemannian manifold $\mathcal{M}$ and let $f : \mathcal{M} \to \mathbb{R}$ be differentiable in a neighborhood of $\mathcal{S}$. Then, the restriction $f|_{\mathcal{S}} : \mathcal{S} \to \mathbb{R}$ is geodesically convex iff for all geodesic segments $c : [0,1] \to \mathcal{M}$ contained in $\mathcal{S}$, we have (for $x = c(0)$)

$$f(c(t)) \geq f(x) + t\langle \mathrm{grad} f(x), c'(0)\rangle_x \qquad \forall t \in [0,1] \tag{1.19}$$

Theorem 1.24. If f is differentiable and geodesically convex on an open geodesically convex set, then x is a global minimizer of f iff $\mathrm{grad} f(x) = 0$.

1.4 Euclidean and Non-Euclidean geometries

In general, in geometry, we say that two lines are parallel iff they are disjoint. In Euclidean geometry, if L is a line and if P is a point not in L, then there exists one and only one line through P that is parallel to L. However, in a hyperbolic geometry, there exist infinitely many distinct hyperbolic lines through P that are parallel to L. Conversely, in elliptic geometries, there exist no line through P that is parallel to L.

The curvature of a surface is defined by the biggest circle that stays on one side of the curve.

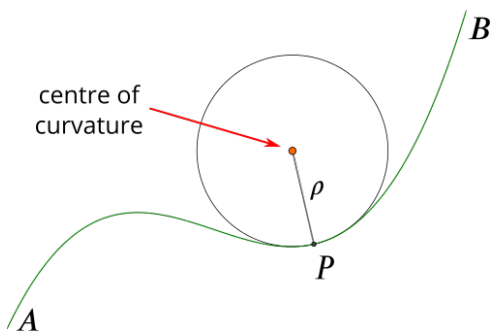

Figure 1.3: Illustration of the curvature and centre of curvature for a function.

The directions in the normal plane where the curvature takes its maximum and minimum values (k_1 and k_2) are always perpendicular and are called principal directions. The product $k_1 k_2$ of the two principal curvatures is called the Gaussian curvature.

Riemannian gradient descent

2.1 Topology tools

2.1.1 General topology

Definition 2.1 (Topology). A topology on a set X is a collection T of subsets of X, called open sets, such that

- X and $\emptyset$ belong to X;
- the union of the elements of any subcollection of T is in T;
- the intersection of the elements of any finite subcollection of T is in T.

Definition 2.2 (Hausdorff topological space.). A topological space is a couple (X, T) where X is a set and T is a topology on X. The topological space (X, T) is Hausdorff if any two distinct points of X have disjoint neighborhoods. If X is Hausdorff, then eveery sequence of points of X converges to at most one point in X.

2.1.2 Topology for embedded submanifolds

Definition 2.3 (Open set.). A subset $\mathcal{U}$ of $\mathcal{M}$ is open (respectively closed) if it is the intersection of $\mathcal{M}$ with an open (respectively closed) set of $\mathcal{E}$.

Definition 2.4 (Neighborhood.). A neighborhood of x in $\mathcal{M}$ is an open subset of $\mathcal{M}$ that contains x.
A neighborhood of a subset of $\mathcal{M}$ is an open subset of $\mathcal{M}$ that contains that subset.

Definition 2.5 (Limit.). Consider a sequence s of points $x_0, x_1, x_2, \ldots$ on a manifold $\mathcal{M}$.

- We way that a point x in $\mathcal{M}$ is a limit of s if, for all neighborhood $\mathcal{U}$ of x, there exists some $K \in \mathbb{Z}$ such that x_k is in $\mathcal{U}$ for all $k \geq K$.
  The topology of a manifold is Hausdorff, hence a sequence has at most one limit. If x is the limit, we write

$$\lim_{k \to \infty} x_k = x \qquad \text{or} \qquad x_k \to x \tag{2.1}$$

  and we say that the sequence converges to x.

- A point $x \in \mathcal{M}$ is an accumulation point of s if it is the limit of a subsequence of s.

2.2 Riemannian gradient descent algorithm

The basic algorithm is the following:

Algorithm 1 Riemannian gradient descent

Input: $x_0 \in \mathcal{M}$;
for $k = 0, 1, 2, \ldots$ **do**
    Pick a step size $\alpha_k > 0$;

$$x_{k+1} = R_{x_k}\left(-\alpha_k \mathrm{grad} f(x_k)\right) \tag{2.2}$$

end for

There are several ways to choose the step size α_k:

- a fixed step size: $\alpha_k = \alpha$ for all k;
- optimal step size: compute α_k that minimizes exactly the function

$$g(\alpha) = f\left(R_{x_k}(-\alpha \mathrm{grad} f(x_k))\right) \tag{2.3}$$

- Backtracking: starting with a guess $\alpha_0 > 0$, iteratively reduce it by a factor $\tau \in (0, 1)$ until it is deemed acceptable.

2.2.1 Global convergence

Let $f : \mathcal{M} \to \mathbb{R}$ be a smooth function on a Riemannian manifold $\mathcal{M}$, and assume that

- there exists a lower bound $f_{low} \in \mathbb{R}$ such that $f(x) \geq f_{low}$ for all $x \in \mathcal{M}$;
- there exists a constant $c > 0$ such that, for all $k \in \mathbb{Z}$,

$$f(x_k) - f(x_{k+1}) \geq c \|\mathrm{grad} f(x_k)\|_{x_k}^2 \tag{2.4}$$

Then,

$$\lim_{k \to \infty} \|\mathrm{grad} f(x_k)\|_{x_k}^2 = 0 \tag{2.5}$$

Furthermore, for all $K \geq 1$, there exists $k \in \{0, \ldots, K-1\}$ such that

$$\|\mathrm{grad} f(x_k)\|_{x_k} \leq \sqrt{\frac{f(x_0) - f_{low}}{cK}} \tag{2.6}$$

This means that if a precision ϵ is required on the solution, the number of iterations that we will need has a complexity $\mathcal{O}(\epsilon^{-2})$.

The proof of these results is the same as done in the courses LINMA2471 or LINMA2460.

2.2.2 Ensuring the assumptions

In classical optimisation, the Lipschitz-smoothness assumption is that there exists a constant $L > 0$ such that

$$\|\nabla f(x+s) - \nabla f(x)\| \leq L\|s\| \qquad \forall x, s \in \mathbb{R}^d \tag{2.7}$$

implied by the expression

$$f(x+s) \leq f(x) + \langle \nabla f(x), s\rangle + \frac{L}{2}\|s\|^2 \qquad \forall x, s \in \mathbb{R}^d \tag{2.8}$$

However, on manifolds, the gradient is defined differently and with a different support. We need to modify this condition.
On a manifold $\mathcal{M}$, for a given subset S of the tangent bundle $\mathcal{TM}$, there exists a constant $L > 0$ such that, forall $(x, s) \in S$,

$$f(R_x(s)) \leq f(x) + \langle \mathrm{grad} f(x), s\rangle + \frac{L}{2}\|s\|_x^2 \tag{2.9}$$

From this, we can ensure the assumption of equation (2.4): define $x =: x_k$, $s =: -\alpha_k \mathrm{grad} f(x_k)$ and $R_x(s) =: R_x(s)$. Then,

$$f(x_{k+1}) \leq f(x_k) - \alpha_k \left(1 - \frac{L\alpha_k}{2}\right) \|\mathrm{grad} f(x_k)\|_{x_k}^2 \tag{2.10}$$

and this gives

$$f(x_k) - f(x_{k+1}) \geq \alpha_k \left(1 - \frac{L\alpha_k}{2}\right) \|\mathrm{grad} f(x_k)\|_{x_k}^2 \tag{2.11}$$

Therefore, the assumption is true taking $c = \alpha_k \left(1 - \frac{L\alpha_k}{2}\right)$.

2.2.3 Optimal step size

From the previous proof, we know that the decrease of the iterate is best when the constant c is maximized. We can thus define the function

$$g : \mathbb{R} \to \mathbb{R} : \alpha \to \alpha \left(1 - \frac{L\alpha}{2}\right) \tag{2.12}$$

which is maximized for $\alpha^* = 1/L$, the same value seen for optimization in $\mathbb{R}^d$. For that value, the constant is $c = 1/2L$.

2.2.4 Backtracking

If the Lipschitz constant L is unknown, a possible algorithm for the optimal step size if backtracking line-search using the Armijo-Goldstein condition. This condition is

$$f(x) - f(R_x(-\alpha \mathrm{grad} f(x))) \geq r\alpha \|\mathrm{grad} f(x)\|_x^2 \tag{2.13}$$

and the algorithm is

Algorithm 2 Backtracking line-search

Parameters: $\tau, r \in (0,1)$, e.g. $\tau = 1/2$ and $r = 10^{-4}$;
Input: $x \in \mathcal{M}$ and $\bar{\alpha} > 0$;
Set $\alpha \leftarrow \bar{\alpha}$;
while $f(x) - f(R_x(-\alpha \mathrm{grad} f(x))) < r\alpha \|\mathrm{grad} f(x)\|^2$ **do**
    $\alpha \leftarrow \tau\alpha$
end while
return α

Under an assumption similar to the Lipschitz condition on the gradient, a convergence rate of $\mathcal{O}(1/\epsilon^2)$ can still be guaranteed.

PAA

3.1 Dynamical systems

Dynamical systems in physics can generally be modelled by a set of differential equations, giving a state-space model:

$$\begin{aligned} \dot{x}(t) &= f(x(t), u(t)) \\ y(t) &= g(x(t), u(t)) \end{aligned} \tag{3.1}$$

where $x \in \mathbb{R}^n$ is the state, $u \in \mathbb{R}^{n_i}$ is the input and $y \in \mathbb{R}^{n_o}$ is the observed output. As their expressions are governed by the system, x and y cannot be free and are restricted to a low-dimensional submanifold in their respective functional space.

3.1.1 The discrete linear time invariant model

This model is

$$\begin{aligned} x(t+1) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \end{aligned} \tag{3.2}$$

Our goal is to determine the dimension n of the state-space model, and the matrix parameters A, B, C, D.
For that, let us define two Hankel matrices:

$$\begin{aligned} Y &= \begin{bmatrix} y(1) & y(2) & \dots & y(N) \\ y(2) & y(3) & \dots & y(N+1) \\ \vdots & \vdots & \ddots & \vdots \\ y(N) & y(N+1) & \dots & y(2N-1) \end{bmatrix} \in \mathbb{R}^{n_o N \times N} \\ U &= \begin{bmatrix} u(1) & u(2) & \dots & u(N) \\ u(2) & u(3) & \dots & u(N+1) \\ \vdots & \vdots & \ddots & \vdots \\ u(N) & u(N+1) & \dots & u(2N-1) \end{bmatrix} \in \mathbb{R}^{n_i N \times N} \end{aligned} \tag{3.3}$$

From the system, we know that

$$\begin{aligned} y(1) &= Cx(1) + Du(1) \\ y(2) &= Cx(2) + Du(2) = C(Ax(1) + Bu(1)) + Du(2) \\ &\vdots \end{aligned} \tag{3.4}$$

By this recurrence, we can see that

$$\begin{bmatrix} y(1) \\ \vdots \\ y(r) \end{bmatrix} = \underbrace{\begin{bmatrix} C \\ CA \\ \vdots \\ CA^{r-1} \end{bmatrix}}_{=:G} x(1) + \underbrace{\begin{bmatrix} D & & & & \\ CB & D & & & \\ CAB & CB & D & & \\ \vdots & \ddots & \ddots & \ddots & \\ CA^{r-2}B & CA^{r-3}B & \dots & CB & D \end{bmatrix}}_{=:H} \begin{bmatrix} u(1) \\ \vdots \\ u(r) \end{bmatrix} \tag{3.5}$$

And thus

$$Y = GX + HU \tag{3.6}$$

where $X = [x(1), x(2), \dots, x(N)] \in \mathbb{R}^{n \times N}$. Let $U^{\perp}$ span the columns of the nullspace of U. Then, $UU^{\perp} = 0$, and

$$YU^{\perp} = GXU^{\perp} + HUU^{\perp} = GXU^{\perp} \tag{3.7}$$

We call $YU^{\perp}$ the input-output matrix, and this is always of rank lower than the dimension of the state space: $rank(YU^{\perp}) \leq n$.
Man-made structures and object often display patterns and/or symmetries in their making, meaning that their actual information space has a much lower dimensionality. Therefore, using the matrix $YU^{\perp}$ is much more efficient.

3.1.2 Discretize a continuous system

To discretize a function $x(t) : \mathbb{R} \to \mathbb{R}$ to a function $x_T(t) : \mathbb{N} \to \mathbb{R}$, we use the Sha function:

$$Ш(t) := \sum_{k=-\infty}^{\infty} \delta(t-k) \tag{3.8}$$

With that function, the discretization is straightforward:

$$x_T(t) = x(t) \cdot Ш\left(\frac{t}{T}\right) \tag{3.9}$$

And in the frequency domain:

$$\mathcal{F}(x_T(t)) = \hat{x}(\nu) \star TШ(t\nu) \tag{3.10}$$

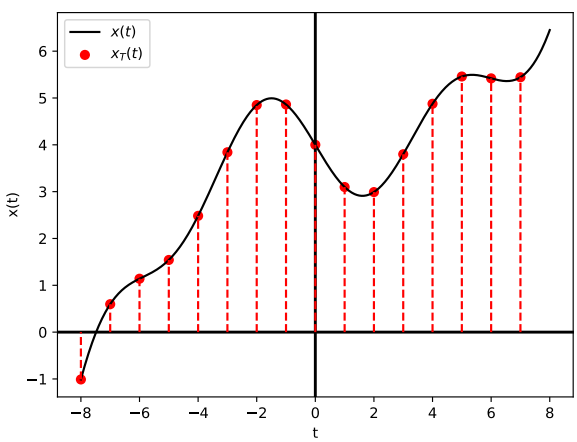

Figure 3.1: Discretization using the Sha function.

3.1.3 Latent variables

A set of random variables $x_o \in \mathbb{R}^{n_o}$ can be given by a probability distribution that depends on other hidden (or latent) variables $x_h \in \mathbb{R}^{n_h}$. In that case, the structure of the data is fully described by the joint distribution of the vector $x = (x_o, x_h) \in \mathbb{R}^n$, for $n = n_o + n_h$. Assuming that $x \sim \mathcal{N}(0, \Sigma)$, where $\Sigma \in \mathbb{R}^{n \times n}$ is the covariance matrix, we define $\Theta = \Sigma^{-1}$ the precision matrix, or concentration matrix. Let us divide the matrices in blocks:

$$\Sigma = \begin{bmatrix} \Sigma_o & \Sigma_{o,h} \\ \Sigma_{o,h}^T & \Sigma_h \end{bmatrix} = \begin{bmatrix} \Theta_o & \Theta_{o,h} \\ \Theta_{o,h}^T & \Theta_h \end{bmatrix}^{-1} \in \mathbb{R}^{n \times n} \tag{3.11}$$

As the matrices of hidden variables are by definition not observed, we need a tool to connect Σ_o to Θ_o. This tool is the Schur complement:

$$\Sigma_o^{-1} = \Theta_o - \Theta_{o,h}\Theta_h^{-1}\Theta_{o,h}^T \in \mathbb{R}^{n_o \times n_o} \tag{3.12}$$

If the observed random variables are mainly independent, then Θ_o is sparse. Moreover, the second term has a rank less than the number of hidden variables n_h, which itself is often relatively small compared to n_o. Therefore, the covariance matrix Σ_o of the observed variables is always a decomposable structure $\Sigma_o^{-1} = S + L$, where S is sparse and L is low-rank, two forms that are very useful in high-dimensional analysis.
Section 1.2.